

Pediatric Myocarditis and Shock



Section I: Scenario Demographics

Scenario Title:	Pediatric viral myocarditis with cardiogenic shock
Date of Development:	January 26, 2017
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY $\geq$ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Adam Cheng, modified by Dawn Lim, Andrea Somers, Nadia Farooki
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To allow learners to practice identifying and managing cardiogenic shock in a critically ill adolescent
CRM Objectives:	1. Demonstrates <i>situational awareness</i> by avoiding fixation error in a dynamic clinical situation 2. <i>Problem solving</i>: considers alternative etiologies of shock when the patient is unresponsive to initial treatment
Medical Objectives:	1. Recognizes a critically ill adolescent with cardiogenic shock 2. Recalls and works through broad differential diagnosis of persistent hypotension 3. Performs safe intubation 4. Manages cardiogenic shock with inotropes

Case Summary: Brief Summary of Case Progression and Major Events

A 15 year-old male with no prior medical history is brought to the ED by his parents for lethargy, shortness of breath and chest pain. He was feeling run down for the past 4 days with URTI symptoms.

His initial presentation looks like sepsis with a secondary bacterial pneumonia. He becomes hypoxic requiring intubation. He develops hypotension that does not respond as expected to fluids and vasopressors, which should prompt more diagnostics from the team.

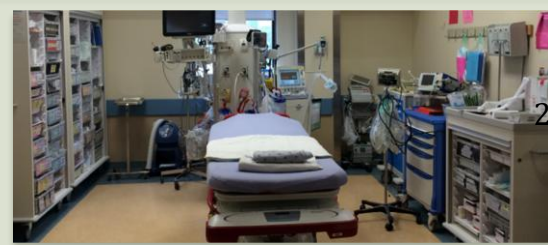
Further testing reveals cardiomyopathy with reduced EF and acute CHF. He finally stabilizes with inotropes and diuresis.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
Up to Date (Pediatric Myocarditis)



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin	Realism:	☒ Conceptual
	☒ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
ED RN	Acts as experienced ED RN		
ED RT	To assist with intubation, ask re: initial vent settings		
B. Required Monitors			
☒ EKG Leads/Wires	☒ Temperature Probe	☒ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☐ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other: Difficult airway cart	
☒ Intraosseous Set-up	☐ LMA	☐ Other: BiPAP machine	
☐ Other: Nebulizer			
D. Moulage			
Adult mannequin sitting-up, gown on with pants, monitor not attached, one IV in place.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

You are working in a large community ED. The charge nurse tells you: "EMS have just arrived with a 15-year old boy with shortness of breath and chest pain. His O₂ sat is low. EMS have administered oxygen and IVF en route. He looks unwell so I put him in a resuscitation room. Can you see him immediately?"

B. Patient Profile and History

Patient Name: Finn O'Connor		Age: 15		Weight: 50 kg	
Gender: ☒ M ☐ F		Code Status: Full			
Chief Complaint: Chest pain and SOB					
History of Presenting Illness: (provided by RN or sim tech) Finn is previously healthy, but has had a viral URTI for the past few days. Many classmates have similar symptoms. He had a fever of 39 degrees yesterday. This morning he complained of increasing shortness of breath and chest pain. His family was concerned about his pallor, lack of energy, and decreased appetite. EMS reported vitals are HR 130, RR 30, O ₂ Sat 84% on room air, BP 110/70.					
Past Medical History:		None		Medications: None	
Allergies: Latex					
Social History: High school student, captain of the basketball team. No history of drug use. No history of mental health issues.					
Family History: None contributory					
Review of Systems:		CNS:		Nil	
		HEENT:		Sore throat	
		CVS:		Chest pain since this morning.	
		RESP:		Short of breath since this morning.	
		GI:		Nil	
		GU:		Nil	
MSK:		Nil.			

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 130/min		BP: 100/70mmHg		RR: 30/min	
O ₂ Sat: 92% on NRB					
Rhythm: Sinus Tach		T: 38 degrees		Glucose: 7.5mmol/L	
GCS: 14 (E3 V5 M6)					
General Status: Toxic looking, lethargic					
CNS:		Drowsy, but responsive			
HEENT:		Erythematous pharynx			
CVS:		Sinus tachycardia, S3, mitral regurg murmur			
RESP:		Bilateral crackles, increased work of breathing, 2-3 word sentences			
ABDO:		Normal			
GU:		Normal			
MSK:		Cool extremities		INT: Pale	

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tachycardia HR: 130/min BP: 110/70 RR: 30/min spO ₂ : 92% NRB T: 38.0°C	Drowsy, but rousable. Laboured breathing. Diaphoretic. Confederate RN to state "did you notice how cold his legs are?"	<u>Learner Actions</u> ☐ History & physical ☐ IV, O ₂ , monitors ☐ Glucose check (7.5) ☐ Appropriate PPE ☐ Broad spectrum Abx ☐ Labs: sepsis/tox/blood Cx/troponin ☐ IV saline bolus ☐ CXR ☐ ECG ☐ ± POCUS	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to the next state</i> All actions complete or 5 min into the case → 2. Hypoxia
2. Hypoxia HR → 140 BP → 90/60 RR → 35 spO ₂ → 82% (slow drop)	Breathing more labored. More drowsy. SpO ₂ starts dropping despite management	<u>Learner Actions</u> ☐ Call RT ☐ Second IV (if not done) ☐ BVM +/- nasal prongs for apneic oxygenation ☐ Airway assessment ☐ Ask for RSI meds ☐ Intubation	<u>Modifiers</u> - If hypoxia is not noticed, RN to say "I think he is becoming more drowsy and his stats are dropping" <u>Triggers</u> Intubation → 3. Hypotension
3. Hypotension HR → 140 BP → 80/50 → 70/40 (automatic drop over 2 min) RR → 12 spO ₂ → 95% (intubated)	Acute drop in BP	<u>Learner Actions</u> ☐ Vasopressor (norepi/phenylephrine) ☐ CXR ☐ POCUS ☐ Verbalize vent settings ☐ Troponin ☐ ECG ☐ Give diuretics ☐ Add inotropic support (<i>dobutamine (0.5-1 mcg/kg/minute) or milrinone (50 mcg/kg over 15 min)</i>) ☐ ± IV corticosteroids ☐ Foley for fluid status ☐ Sedation	<u>Modifiers</u> - If given IV bolus → BP worsens 60/40 - No response to pressors - If no inotrope started 5 min into state, RN to state "What do you think is going on here? Is this just sepsis? What's your differential?" <u>Triggers</u> Inotropes started → 4. Resolution
4. Resolution HR → 110 BP → 100/60 RR → 12 spO ₂ → 95%	Sedated Intubated Ventilated	☐ STAT formal Echo ☐ Consider ECMO/bypass ☐ Call pediatric cardiology ☐ Call pediatric ICU	<u>Modifiers</u> <u>Triggers</u> Call PICU → End case

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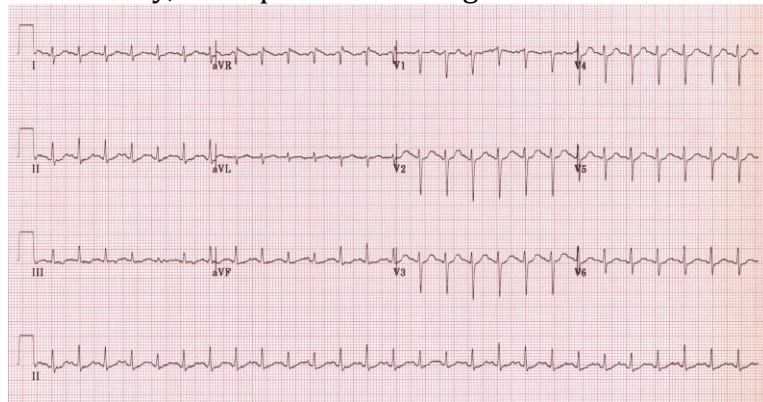


Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results					
VBG	pH: 7.15	PCO ₂ : 48mmHg	PO ₂ : 18mmHg	HCO ₃ : 13mmol	

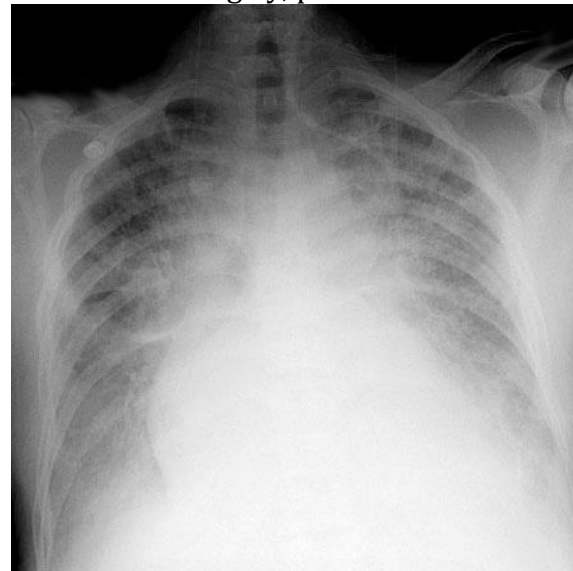
Images (ECGs, CXRs, etc.)

ECG – tachy, non-specific ST changes



<https://lifeinthefastlane.com/ecg-library/myocarditis/>

CXR – cardiomegaly, pulm edema



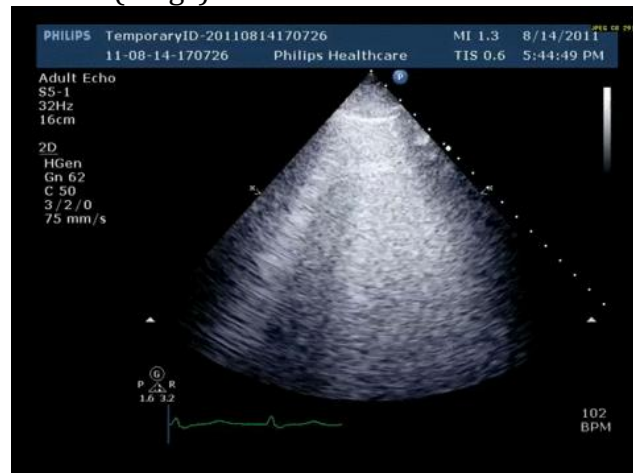
<https://www.med-ed.virginia.edu/courses/rad/cxr/postquestions/posttest.html>

POCUS (parasternal long) – Impaired EF



<http://www.thepocusatlas.com/echo/xg2awokhx1zx8q3ndwju5cu4t1adq>

POCUS (lungs) – B-Profile



<https://www.thoracic.org/professionals/clinical-resources/critical-care/clinical-education/quick-hits/orthopnea-in-a-patient-with-doxorubicin-exposure.php>

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
Medical Knowledge 1. What factors made (or could have made) intubation difficult? Did you assess these prior to attempting intubation? 2. How did you optimize the patient prior to intubation? Would this change if you knew the patient had cardiogenic shock? 3. Describe your differential for post-intubation hypotension. How do you work through this list? 4. How do inotropes work? When would you consider using them in the ED? CRM Skills 1. How can you re-orient when a patient does not respond to treatment as you expect? 2. What resources can you use when facing a diagnostic challenge? 3. Do you (team leader) feel that you utilized your team members effectively? Does the team feel like they were used effectively?	
Key Moments	
Decision to intubate/Intubation planning	
Recognizing myocarditis/cardiogenic shock	
Providing inotropic support/hemodynamic optimization	